

FINAL PROGRESS REPORT – March 30, 2026

Project: CFPB Complaint Analysis (*Data Vintage - Feb 2024 to Feb 2026*)

Aim: Model customer complaint data using NLP and HMM for complaint triage and escalation modeling.

Tasks Accomplished since First Progress Report:

1. Consolidated 5-class company response labels into 3 meaningful resolution tiers representing service complexity levels as target variable.
2. Implemented temporal validation framework with OOS (Feb 2024 - Jul 2025) and OOT (Aug 2025 - Nov 2025) data split.
3. Trained and evaluated three baseline models namely Logistic Regression, Random Forest and XGBoost and concluded XGBoost as the winner model and performed hyperparameter tuning using grid search for it.
4. Tested, validated and generated synthetic links between complaints based on tight linking criteria: same company, same zip code, same topic category, temporal proximity (7-60 days), degrading sentiment (same or worsening sentiment) and a logical tier progression (same or escalating tier).

Current Output: 965K complaints processed into 624K sequences (about 29% multi-complaint chains) for HMM.

Immediate Next Steps:

1. To start inspecting the multi-chains and modelling HMM on those to discover hidden states. First, I will validate HMM implementation on a 10K sequence subset before scaling to full dataset of 624K sequences
2. I will interpret the discovered hidden states and analyze them, followed by escalation pathway scenarios.

Challenges encountered:

1. Hyperparameter Tuning

Challenge : Grid-searched parameters improved OOS performance but showed degradation on OOT set.

Solution : Retained baseline XGBoost configuration prioritizing temporal generalization over OOS optimization. This will favor robustness for real-life situations where future data distribution may shift.

2. Suspected feature dominance due to skewness

Challenge : **dominant_topic** feature single-handedly had more importance than other NLP features.

Solution : Re-ran all models excluding **dominant_topic**. XGBoost remained best performer with other NLP features. I concluded that the model learned meaningful patterns beyond this feature.

3. Tightening synthetic link criterion

Challenge : Required a robust criterion to create meaningful and logical synthetic links to model HMM.

Solution : Tried with a simple criterion first to try simulating links on a base of 10K. Tightened the criterion with hard conditions related to zip codes, sentiment and tier escalation. Evaluated this for a sample of 100K and scaled to full dataset upon verifying the results.

Overall plan for final submission:

Substantial progress was achieved since first report with core supervised learning pipeline complete and validated. I also completed the synthetic sequence generation for full dataset, enabling the HMM phase. Now, my remaining work includes modelling HMM, analyzing the hidden states and identifying escalation pathways. Finally, I will integrate the results from earlier phases (NLP and HMM). With the current momentum, my project would maintain the decided scope of addressing both complaint triage (supervised) and escalation modeling (HMM) objectives. The plan is to complete HMM on full 624K sequences, with a 10K subset serving as initial validation. If time constraints arise, the validated 10K implementation will still demonstrate methodology feasibility. I also intend to work on the documentation and paper writing part of the project in the upcoming week to stay on track with the project deadline. The final expected outcome is a dataset rich in NLP features and modeled hidden states capable of identifying the escalation pathways in customer complaints. This project will provide a probabilistic analysis of customer complaints getting escalated in a given scenario. For the current scope of the project's final submission, I plan to wrap up with HMM and conclude with the technical paper, although future work could include validation with actual customer sequences if IDs become available and A/B testing of intervention strategies. This methodology can also be expanded to other financial service complaint datasets.